A Multigranularity Spatiotemporal Attention Model Based on Multisource Satellite Data for Monthly XCO_2 Reconstruction Over China

Ruizhi Chen, Mingmin Zou, Cuihong Chen, Huizhen Xie, Chunyan Zhou, Huiqin Mao, and Zhongting Wang

***Abstract*—Accurate and timely monitoring of atmospheric carbon dioxide (CO_2) concentrations is a crucial prerequisite for achieving carbon peaking and carbon neutrality goals. However, existing satellite observations are often limited by cloud cover, sensor constraints, and orbital characteristics, resulting in data gaps that hinder the direct analysis of CO_2 spatiotemporal dynamics. To address this, we develop an operational processing pipeline that leverages an integrated deep learning framework to reconstruct monthly, full-coverage column-averaged CO_2 dry-air mole fraction (XCO_2) over China at 0.05° spatial resolution for the period 2019–2024. This pipeline integrates multisource satellite data [Greenhouse gases Observing Satellite (GOSAT), Orbiting Carbon Observatory-2 (OCO-2), and Gaofen-5B (GF-5B)] with auxiliary variables, including vegetation indices, nighttime light remote sensing, NO_2 column concentrations, and fifth-generation ECMWF reanalysis dataset (ERA5) meteorological parameters. By eliminating reliance on assimilated model products, the pipeline generates monthly XCO_2 maps with a processing latency of 1–2 months after satellite observation. Validation against total carbon column observing network (TCCON) ground-based observation demonstrates strong reconstruction performance (R^2 = 0.88 and root-mean-squared error (RMSE) = 1.50 ppm). When compared to CarbonTracker (CT) and copernicus atmosphere monitoring service (CAMS) datasets, the reconstructed dataset shows mean biases of −0.28 and −0.46 ppm, and standard deviations of 0.53 and 0.72 ppm, respectively, indicating that it accurately captures the spatial distribution and interannual trends of CO_2 concentrations. Comparative experiments further reveal that incorporating GF-5B data improve the overall model performance. Regional analyses show that the model effectively identifies high-emission hotspot areas. Interannual analysis indicates that the average XCO_2 in China increased from 410.30 ppm in 2019 to 422.73 ppm in 2024, corresponding to an annual growth rate between 0.4% and 0.9%.**

***Index Terms*—Deep learning, high-resolution column-averaged CO_2 dry-air mole fraction (XCO_2), multisource satellite data, spatiotemporal reconstruction.**

I. Introduction

CARBON dioxide (CO_2) can absorb longwave radiation emitted from the Earth's surface, contributing to the greenhouse effect and leading to a rise in the planet's average temperature. This warming trend has already caused serious consequences, including sea-level rise and an increased frequency of extreme weather events [1]. Under natural conditions, atmospheric CO_2 concentration tends to remain relatively stable. However, due to intensive emissions from human activities such as energy production, industrial operations, and transportation [2], the global average CO_2 concentration has increased from 336.85 ppm in 1979 to 422.81 ppm in February 2024 [3]. The rapid increase in CO_2 levels has drawn significant international attention. Governments worldwide have signed various agreements aimed at controlling greenhouse gas emissions. Notably, the 2015 Paris Agreement set a target to limit global temperature rise to below 2 °C, with efforts to constrain it within 1.5 °C [4]. As the largest emitter of carbon globally, China has made substantial commitments to mitigate emissions through its "dual-carbon" strategy, which aims to peak carbon emissions by 2030 and achieve carbon neutrality by 2060 [5].

Currently, there are three main approaches for obtaining atmospheric CO_2 concentrations, depending on the observation platform: ground-based measurements, airborne measurements, and satellite remote sensing [6]. Ground-based systems, such as those in the total carbon column observing network (TCCON), provide highly accurate and temporally continuous measurements of near-surface or column-integrated CO_2 concentrations [7]. However, the limited spatial distribution of monitoring stations restricts their ability to support large-scale continuous observations. Airborne platforms offer greater flexibility but are constrained by high operational costs and limited flight coverage, which hinders their capability to provide high-frequency, large-area data [8]. In contrast, satellite remote sensing has become widely used for acquiring CO_2 concentration data at both global and regional scales due to its ability to deliver continuous observations with high spatial resolution [9]. Nonetheless, satellite observations are subject to limitations such as cloud cover, sensor performance, and orbital constraints, which can result in gaps in data coverage.

In the past decade, machine learning theories and applications have advanced rapidly, particularly in the domain of

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Ruizhi Chen is with North China University of Water Resources and Electric Power, Zhengzhou 450046, China, and also with Chinese Research Academy of Environmental Sciences, Beijing 100012, China (e-mail: chenruizhi23@mails.ucas.ac.cn).

Mingmin Zou is with Anhui University, Hefei 230039, China (e-mail: zoumm@ahu.edu.cn).

Cuihong Chen, Huizhen Xie, Chunyan Zhou, and Huiqin Mao are with the Satellite Application Center for Ecology and Environment, Ministry of Ecology and Environment, Beijing 100094, China (e-mail: chench_07@163.com; huizhenxc@126.com; mezhouchunyan@126.com; maohq@seecmep.cn).

Zhongting Wang is with North China University of Water Resources and Electric Power, Zhengzhou 450046, China (e-mail: wangzhongting@ncwu.edu.cn).

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reconstructing continuous CO_2 distributions [10], [11]. The increasing availability of high-quality ground-based observations, assimilation products, and satellite remote sensing data has provided a strong foundation for CO_2 reconstruction research [12]. Meanwhile, the ability of machine learning models to extract spatial and temporal features has enabled them to effectively address the complex spatiotemporal heterogeneity inherent in CO_2 distributions [13]. Early studies primarily relied on statistical models that lacked explicit spatiotemporal structure, which often faced limitations when integrating heterogeneous multisource data and capturing complex spatiotemporal dependencies. Siabi et al. [14] applied a multilayer perceptron (MLP) to fuse Orbiting Carbon Observatory-2 (OCO-2) satellite data with eight environmental variables, generating a continuous spatial CO_2 distribution over Iran during the 2015 growing season. Similarly, Li et al. [15] developed an extremely randomized trees (ERTs) model to produce a global 0.01° resolution, eight-day CO_2 dataset based on OCO-2 observations and auxiliary variables, while He et al. [16] utilized the LightGBM algorithm to fill gaps in OCO-2 data across China from 2015 to 2018. To overcome the limitations of traditional models, more recent approaches have incorporated deep learning architectures that capture both spatial and temporal dependencies. For example, Zhang and Liu [17] employed a convolutional neural network (CNN) with a convolutional block attention module (CBAM) to model spatial relationships among meteorological variables, creating a long-term (2003–2020) dataset at 0.25° resolution for China. He et al. [18] further proposed a CNN–long short-term memory (LSTM)–attention model to extract spatiotemporal features simultaneously, reconstructing CO_2 concentrations over China from 2014 to 2022. In addition, some studies have attempted to improve spatial coverage by combining assimilation products with satellite observations through machine learning models. However, such approaches often suffer from temporal latency and are unsuitable for near-real-time applications, as demonstrated in studies by Tian et al. [19] and Yi et al. [20], who applied feedforward neural networks and the deep learning-based spatiotemporal model (DSTM) model to generate daily CO_2 concentration datasets.

While the aforementioned studies have made significant contributions to the field of spatiotemporal reconstruction for column-averaged CO_2 dry-air mole fraction (XCO_2), most rely on model-assimilated data with inherent latency. This limitation hinders the generation of near-real-time datasets. Moreover, although existing models demonstrate capabilities in extracting spatiotemporal features, their reliance on simplistic, single-granularity representations may fail to capture the complex underlying spatiotemporal correlations of CO_2 concentrations. To address these challenges, this study develops an integrated processing pipeline that refines and integrates several classical deep learning modules within a spatiotemporal attention framework, enabling the fusion of multisource satellite CO_2 observations—including the underexplored Gaofen-5B (GF-5B) sensor—with auxiliary variables. By explicitly accounting for the spatial context of satellite footprints, the approach effectively mitigates striping artifacts and produces seamless monthly XCO_2 fields at 0.05° spatial resolution over China for the period 2019–2024. Moreover, the system operates without reliance on lagged data assimilation products, thereby facilitating the near-real-time reconstruction of CO_2 distributions. In terms of accuracy, previous studies have reported root-mean-squared errors (RMSEs) ranging from approximately 1.0 to 1.8 ppm when validated against TCCON ground-based measurements, whereas the proposed model achieves an RMSE of 1.5 ppm under the same validation protocol. Our dataset provides a more accurate depiction of regional scale CO_2 concentration distribution and variability. This enhanced resolution is crucial for detailed analysis of regional carbon dynamics and supports policymakers in establishing routine monitoring systems. Comparisons with ground-based stations and other model-assimilated products demonstrate strong agreement, validating the effectiveness of our proposed method.

II. Data

The data utilized in this study comprise three components: 1) satellite-observed CO_2 concentration data, 2) auxiliary variables, and 3) ground-based station measurements (used for accuracy validation) alongside model simulation data (employed for comparative analysis). Representative spatial distributions for selected variables are illustrated in Fig. 1. The specific parameters of all variables are summarized in Table I.

A. XCO_2 Satellite Data

Satellite-observed CO_2 concentration data serve as the training labels for our model and constitute the most critical input in this study. It is essential to note that while CO_2 is relatively well mixed in the atmosphere, its total vertical column abundance can be significantly influenced by terrain and atmospheric pressure. To ensure comparability across varying conditions, satellite observations typically represent CO_2 using the XCO_2, derived by normalizing the CO_2 vertical column density using oxygen (O_2) measurements. O_2 is employed as the reference gas because its atmospheric concentration is known and remains highly stable. The datasets from three carbon-observing satellites are briefly described in Sections II-A1–II-A3

1) GOSAT and GOSAT-2: Japan launched the Greenhouse gases Observing Satellite (GOSAT) on January 23, 2009. As the first satellite dedicated to greenhouse gas monitoring, GOSAT achieved global-scale observations of atmospheric CO_2 with high spectral resolution. Its onboard Fourier transform spectrometer (FTS) measures reflected solar shortwave radiation and terrestrial thermal emission from Earth's surface. CO_2 concentrations and vertical profiles are retrieved by analyzing the distribution and structure of the observed spectra. GOSAT has an observation footprint diameter of 10.5 km and a revisit cycle of three days. This study utilizes level 2 (L2) XCO_2 products derived from the short-wavelength infrared (SWIR) band, covering the period January 2019 to December 2024 (source: https://data2.gosat.nies.go.jp/index_en.html).

GOSAT-2 was successfully launched on October 29, 2018, and inserted into its planned orbit. It carries an enhanced version of the original GOSAT sensor, the FTS-2 instrument,

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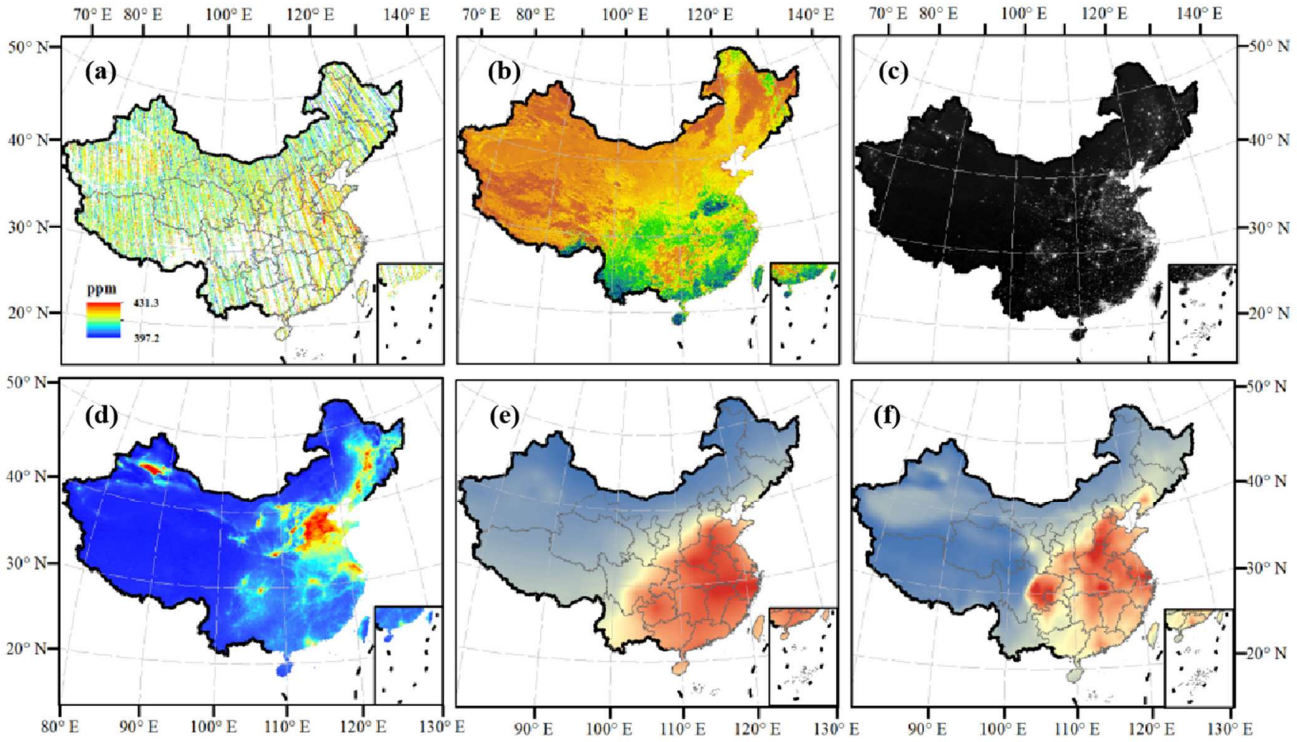


Fig. 1. (a) Representation of XCO_2 data obtained from GOSAT, GOSAT-2, OCO-2, and GF-5B for 2023. (b) Monthly NDVI derived from MODIS observations in January 2023. (c) Monthly cloud-free nighttime light products derived from EOG in January 2023. (d) Monthly mean column amount of NO_2 derived from TROPOMI observations in January 2023. (e) CT XCO_2 data for January 2020. (f) CAMS-EGG4 XCO_2 data for January 2020.

TABLE I
SUMMARY OF DATA USED IN THIS STUDY

Date	Variable	Swath width/Temporal resolution	Spatial resolution	Period
XCO_2	GOSAT XCO_2	790km	10.5km	2019/01 – 2024/12
	GOSAT-2 XCO_2	903km	9.6km	2019/03 – 2024/01
	GF-5B XCO_2	640km	10.3km	2023/01 – 2024/12
	OCO-2 XCO_2	10.6km	2.25×1.29km	2019/01 – 2024/12
Vegetation Index	NDVI EVI	16d	1km	2019/01 – 2024/12
Source-related Pollutants	TROPOMI NO_2	Monthly	3.5×7km	2019/01 – 2024/12
Nighttime Light	SNPP_VIIRS_DNB	Monthly	0.025°	2019/01 – 2024/12
ERA5 meteorology	2-meter air temperature Boundary layer height Surface pressure Total precipitation U wind component V wind component Total column water vapor Skin temperature	Monthly	0.25°	2019/01 – 2024/12

featuring a larger field of view and a higher signal-to-noise ratio (SNR). GOSAT-2 has a footprint diameter of 9.7 km and a revisit cycle of six days [21]. We employed the thermal and near-infrared sensor for carbon observation (TANSO)-FTS-

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2 L2 short-wavelength full physics (SWFP) product (source: https://prdct.gosat-2.nies.go.jp/index.html.en), with data available from March 2019 to January 2024.

2) OCO-2: The OCO-2, launched by the National Aeronautics and Space Administration (NASA) in 2014, is a dedicated greenhouse gas monitoring satellite. It carries a high-resolution grating spectrometer with three distinct channels: the O_2 A-band, the weak CO_2 absorption band, and the strong CO_2 absorption band. The instrument achieves a spectral resolution of up to 0.04 nm, with a spatial resolution of 1.29 $\times$ 2.25 km and a revisit cycle of 16 days [22].

3) GF-5B: The GF-5B is the second satellite to achieve hyperspectral resolution observation in China High-Resolution Earth Observation System (CHEOS), which was launched in September 2021. The greenhouse gases monitoring instrument (GMI) is one of seven payloads onboard GF-5B. It is designed with four bands covering 765–769, 1568–1583, 1642–1658, and 2043–2058 nm, and mainly used to obtain global CO_2 and methane column abundance [23]. Estimate of the XCO_2 is retrieved using optimal estimation (OE) algorithm along with machine learning method, with accuracy better than 0.4% [24]. GMI/GF-5B XCO_2 data from 2023 to 2024 are used in this study.

B. Auxiliary Data

Regional atmospheric CO_2 concentrations are influenced by a variety of factors, including meteorological conditions, vegetation dynamics, and anthropogenic activities, as well as the complex interactions among them. The underlying mechanisms of these influences have been investigated in previous studies [25], [26]. In this study, a diverse set of variables is utilized, including vegetation indices, meteorological reanalysis data, co-emitted trace gases, and nighttime light remote sensing data. The details of these variables are described as follows.

Terrestrial vegetation plays a pivotal role in the carbon cycle by regulating the exchange of CO_2 between the land surface and the atmosphere through photosynthesis and respiration. Thus, variations in vegetation cover and activity directly influence atmospheric CO_2 concentrations. To capture these dynamics, vegetation indices are commonly employed, as they integrate spectral information that is sensitive to canopy greenness, photosynthetic activity, and biomass [27]. Therefore, this study employs the normalized difference vegetation index (NDVI) and the enhanced vegetation index (EVI) to represent the influence of terrestrial natural carbon sinks on regional CO_2 concentrations. Specifically, the MOD13A2 product from the Moderate Resolution Imaging Spectroradiometer (MODIS) is used (available at: https://lpdaac.usgs.gov/products/mod13a2v061/), which provides 1-km spatial resolution data at a 16-day interval.

Previous studies have demonstrated the feasibility of using co-emitted trace gases to estimate regional carbon emissions [28]. Nitrogen dioxide (NO_2), which is closely associated with anthropogenic activities—particularly fossil fuel combustion and biomass burning—was selected as one of the auxiliary variables in this study to represent the contribution of anthropogenic sources to CO_2 concentrations. Monthly NO_2 concentration grids at 1-km spatial resolution from January 2019 to December 2024 were generated using the NRTI/L3_NO_2 product (available at: https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S5P_NRTI_L3_NO2#description) via the Google Earth Engine (GEE) platform.

Anthropogenic activities, such as fossil fuel combustion and industrial production, constitute the dominant sources of atmospheric CO_2 emissions. These activities are closely associated with urbanization and energy consumption, both of which can be effectively represented by satellite-derived nighttime light data. Nighttime light intensity provides a reliable proxy for the spatial distribution and intensity of human activities, reflecting population density, economic development, and energy use. Elvidge et al. [29] and Doll et al. [30] demonstrated a correlation between nighttime light brightness and carbon emissions, suggesting its applicability in estimating and modeling regional anthropogenic CO_2 emissions. In this study, the monthly cloud-free day/night band (DNB) composite products released by the Earth Observation Group (EOG) were utilized (available at: https://eogdata.mines.edu/products/vnl). Specifically, the "vcmsl" version was selected, which provides a spatial resolution of 15 arc-second (approximately 500 m at the equator).

Meteorological parameters are indispensable variables in characterizing the carbon cycle process [10]. Temperature, precipitation, radiation, and boundary layer dynamics directly influence both natural and anthropogenic components of the carbon budget. For instance, temperature and solar radiation regulate photosynthesis and respiration, while precipitation affects vegetation growth and soil moisture conditions. In addition, wind speed and boundary layer height determine the dispersion and vertical mixing of atmospheric CO_2. Consequently, meteorological parameters provide essential constraints on both biospheric carbon fluxes and the transport of CO_2 in the atmosphere, and are therefore included in this study as critical auxiliary variables. In this study, the fifth-generation ECMWF reanalysis dataset (ERA5) was employed. ERA5 integrates observational data and model estimates through advanced data assimilation techniques, making it one of the most accurate and comprehensive reanalysis products currently available [32]. It provides hourly global meteorological variables at a spatial resolution of 0.25°. Seven key variables were selected for this study: 2-m air temperature, boundary layer height, surface pressure, total precipitation, 10-m v-component of wind, 10-m u-component of wind, skin temperature, and total column water vapor. Detailed descriptions and units of each variable can be found in the official documentation (https://cds.climate.copernicus.eu/datasets/reanalysis-era5-complete?tab=overview).

C. Ground-Based Observations and Model Assimilation Data

Ground-based observations offer high-precision and temporally continuous measurements of atmospheric CO_2 concentrations, while high-quality model assimilation products provide reliable estimates of spatial CO_2 distributions [13]. To assess the accuracy and robustness of the reconstructed dataset presented in this study, measurements from the TCCON were used for validation. In addition, model-based assimilation

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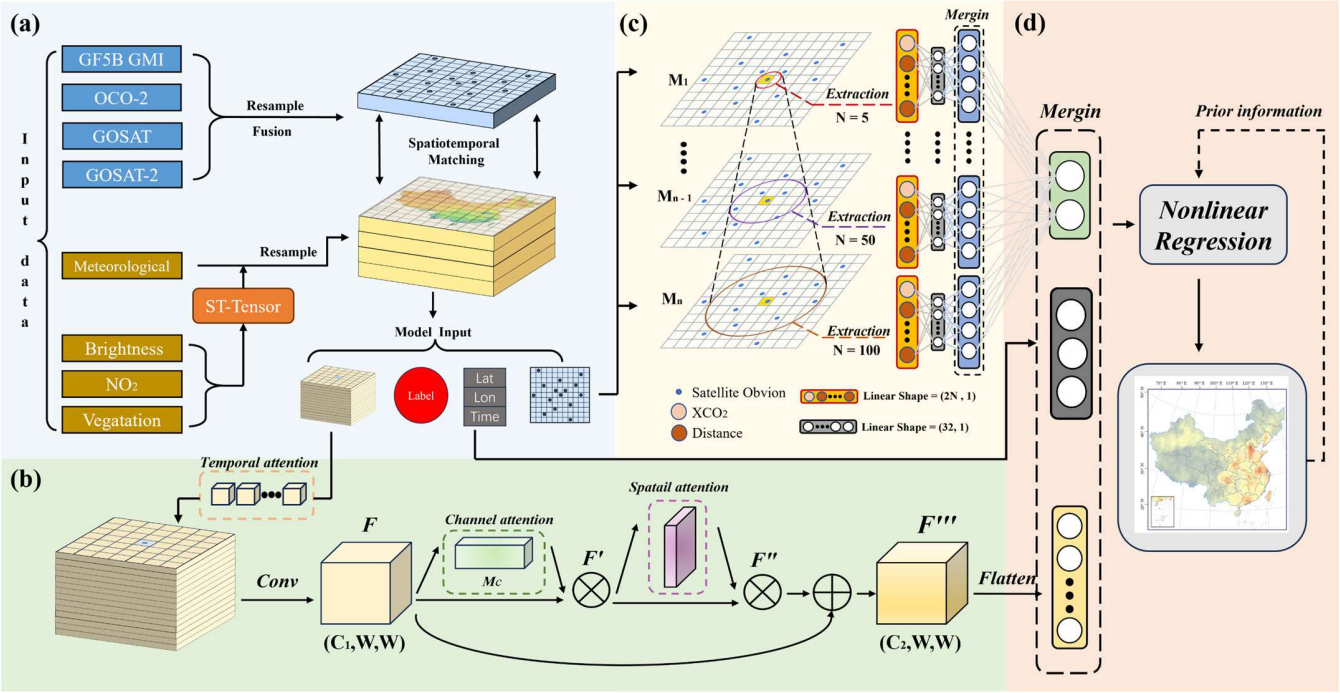


Fig. 2. Dataflow of the study. (a) Data preprocessing and spatiotemporal matching. (b) TSCAM. (c) XCO_2 multigranularity feature extraction module. (d) Feature integration and nonlinear regression.

products, including CarbonTracker (CT) and copernicus atmosphere monitoring service (CAMS)-global greenhouse gas reanalysis (EGG4), were employed for comparative evaluation. A brief description of these datasets and their respective data sources is provided as follows.

The TCCON employs high-resolution FTSs to measure the XCO_2 across the entire atmosphere [33]. Compared to near-surface measurements, XCO_2 data are less affected by vertical mixing and local emissions, offering higher representativeness and accuracy, which makes them widely used for validating satellite retrievals. In China, two TCCON stations are operational: Hefei (since 2015) and Xianghe (since 2018), both providing continuous XCO_2 observations. Given that the reconstructed dataset in this study is gridded at a 0.05° spatial resolution and monthly temporal resolution, we selected measurements from 13:00 to 14:00 local time and aggregated them into monthly averages to ensure spatiotemporal consistency for validation purposes.

CT, developed by the National Oceanic and Atmospheric Administration (NOAA), is an atmospheric CO_2 data assimilation system based on the TM5 transport model, incorporating observations from world meteorological organization (WMO) global background stations [34]. It has been widely used for satellite data validation, atmospheric model forcing, and carbon flux analysis. This study employs the CT2022 release (https://gml.noaa.gov/ccgg/carbontracker/), which provides data at a 3-h temporal resolution and a $2° \times 3°$ spatial resolution, covering the period from January 2019 to December 2020. CAMS-EGG4, developed under the CAMS, assimilates CO_2 retrievals from satellites such as SCIAMACHY, GOSAT, and IASI [35]. It offers a three-hourly dataset at a 0.75° spatial resolution (https://ads.atmosphere.copernicus.eu/datasets/cams-global-ghg-reanalysis-egg4?tab=download), and is one of the most widely used global reanalysis products for greenhouse gases. Given that satellite overpass times—such as those for GOSAT and OCO-2—are typically concentrated between 13:00 and 13:30 local time, we averaged the CT and CAMS-EGG4 outputs at 12:00 and 15:00 to produce reference datasets more representative of the satellite observation window.

III. Math

Fig. 2 presents the workflow of this study. The process begins with the standardization of input data formats, followed by the division of the dataset into training and testing subsets for model development. Subsequently, model training and hyperparameter tuning are conducted to optimize performance. Finally, the model is evaluated for accuracy and applied to generate the spatiotemporally reconstructed gridded XCO_2 dataset.

A. Data Preprocessing

The data preprocessing workflow, as illustrated in Fig. 2(a), consists of three main components: the fusion of multisource satellite observations, the gap-filling and resampling of auxiliary variables, and the spatiotemporal alignment for generating training samples. The specific procedures are described as follows.

1) Fusion of Satellite Data: Due to the differing sensor configurations aboard various satellites, each instrument exhibits varying sensitivities to CO_2 concentrations across different atmospheric layers. Therefore, prior to fusing satellite observations from multiple sources, their column averaging

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kernels must be harmonized. Following the approach proposed by He et al. [36], this study applies a correction to the satellite data using prior vertical CO_2 profiles provided by CT, as described in (1). Since CT profiles are only available up to December 2020, pressure profiles from ERA5 reanalysis data were adopted for kernel adjustment after this period

$$\mathrm{XCO}_{2_{\mathrm{adj},t}} = \mathrm{XCO}_{2_{\mathrm{ret},t}} + \boldsymbol{h}^{\mathrm{T}} \cdot (\boldsymbol{I} - \boldsymbol{A}) \cdot \left(X_{M,t} - X_{a,t}\right) \tag{1}$$

where $\mathrm{XCO}_{2_{\mathrm{adj},t}}$ denotes the column-averaged CO_2 concentration after averaging kernel adjustment, while $\mathrm{XCO}_{2_{\mathrm{ret},t}}$ presents the original retrieved XCO_2. The variable $\boldsymbol{h}$ is the pressure weighting vector. $\boldsymbol{I}$ denotes the identity matrix. $\boldsymbol{A}$ refers to the column averaging kernel provided by the satellite retrieval algorithm. $X_{M,t}$ is the prior CO_2 profile from CT. $X_{a,t}$ is the satellite-specific prior profile used in the retrieval process.

After adjusting the column averaging kernels, monthly fusion and gridding operations were applied to the XCO_2 data from multiple satellites, resulting in $0.05^\circ \times 0.05^\circ$ spatial resolution monthly averaged XCO_2 raster datasets. The fusion process considers three possible cases: 1) if no satellite observations fall within a given pixel, the pixel is assigned a value of "no data"; 2) if only a single observation falls within a pixel, the value is directly assigned to the pixel; and 3) if multiple observations fall within the same pixel, the pixel value is computed using a weighted average, where the inverse of the retrieval uncertainties serves as the weights. The calculation is given in the following equation:

$$\mathrm{XCO}_2^{(r)} = \frac{\sum_k \frac{1}{\omega_k^2} \mathrm{XCO}_2^{(k)}}{\sum_k \frac{1}{\omega_k^2}} \tag{2}$$

where $\mathrm{XCO}_2^{(r)}$ denotes the fused XCO_2 value for a given pixel, $\mathrm{XCO}_2^{(k)}$ represents the k-indexed individual satellite observation within that pixel, and ω_k is the uncertainty associated with the k-indexed satellite observation.

2) Auxiliary Data Processing: To enable accurate spatiotemporal alignment with the fused XCO_2 grid, all auxiliary datasets were standardized in terms of spatial and temporal resolution. The following preprocessing steps were performed.

1) To address missing values in NDVI, EVI, NO_2, and nighttime light data, the spatial–temporal (ST)-tensor algorithm [37] was applied. This method jointly optimizes spatiotemporal kernel functions and a low-rank tensor structure to ensure that the imputed values preserve the inherent spatiotemporal continuity of geophysical parameters.
2) Parameters exhibiting power law distributions, such as TROPO spheric monitoring instrument (TROPOMI) NO_2 and suomi national polar-orbiting partnership (SNPP)_visible infrared imaging radiometer suite (VIIRS)_DNB brightness values, were log-transformed to reduce skewness and stabilize variance.
3) For NDVI and EVI, originally available at 16-day intervals, the corresponding values were aggregated to monthly means.
4) All auxiliary data were resampled to a 0.05° spatial resolution using bilinear interpolation.
5) To eliminate differences in data scale and improve model convergence, all auxiliary variables were standardized using z-score normalization, as defined in the following equation:

$$x_s = \frac{x - \bar{x}}{x_{\mathrm{std}}} \tag{3}$$

where x_s denotes the standardized value of the auxiliary variable, x represents the original input, $\bar{x}$ is the mean of the original input, and x_{std} is the standard deviation of the original input.

3) Training Data Generation: After standardization, all multisource input data were aligned to a uniform spatiotemporal resolution. The training data matrix was constructed using a combination of data cube stacking and spatial cropping techniques, comprising the following four structured components.

1) *Auxiliary Feature Tensor:* For each target XCO_2 grid cell, a spatial window of size $W \times W$ centered on the pixel is extracted. Within this window, C-channel auxiliary environmental variables are extracted. By aggregating the data over T consecutive months, a sequence of auxiliary feature tensors is formed, resulting in a 4-D tensor $X_{\mathrm{aux}} \in R^{T \times C \times W \times W}$ (with $W = 15$, $C = 12$, and $T = 5$ in this study). This tensor captures rich multiscale spatiotemporal contextual information centered on the target location.
2) *Neighborhood XCO_2 Observation Sequence:* Based on spatial proximity, the N nearest valid XCO_2 observations around the target pixel ($N = 100$ in this study) are selected to construct a feature vector $X_{\mathrm{neighbor}} \in R^{2N \times 1}$. Each pair of elements in this vector consists of the XCO_2 value of a neighboring observation and its relative spatial distance to the target pixel.
3) *Spatiotemporal Encoding Vector:* The temporal information (i.e., month index) and the geographical coordinates (latitude and longitude) of each target XCO_2 pixel are extracted to form a spatiotemporal feature vector $X_{\mathrm{st}} \in R^{3 \times 1}$. To better capture seasonal periodicity, the temporal component is encoded using a sine function. This encoding approach allows the model to effectively learn cyclical temporal patterns in the data, as described in the following equation:

$$t_{\mathrm{enc}} = \left(\frac{t}{12}\right) \cdot \sin\left(\frac{2\pi t}{12}\right) \tag{4}$$

where t_{enc} denotes the encoded temporal value and t represents the month index counted sequentially from January 2019 to December 2024.

After the aforementioned processing steps, approximately 1.5 million valid data samples were obtained. To reduce computational costs, stratified sampling was performed across different months, yielding a total of 300 000 representative samples. These samples were subsequently split into training and testing sets in a 3:1 ratio.

B. Multigranularity Spatiotemporal Attention Network

1) Temporal–Spatial Channel Attention Module: Fig. 2(b) illustrates the architecture of the temporal–spatial channel attention module (TSCAM), a lightweight attention mechanism that integrates temporal, spatial, and channelwise information. Following the approach proposed in [38], this

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module extends the CBAM by incorporating an MLP to compute attention weights along the temporal dimension. In addition, a residual connection is introduced at the end of the module to preserve feature integrity. The core idea behind TSCAM is to jointly model temporal–spatial dependencies while dynamically reweighting feature channels, thereby enhancing the model's ability to focus on salient information.

When an auxiliary feature tensor sequence with the shape of (T, C, W, W) is fed into TSCAM, the temporal attention module first computes attention weights across the T time steps and performs weighted fusion of the multitemporal features, generating a temporally representative feature base. This is followed by two sets of 3×3 convolutional layers (stride = 1 and padding = 1) to enhance spatial representations, resulting in an intermediate feature map $F \in R^{C_1 \times W \times W}$. To enhance the representation capability of the feature map F, we adopt the CBAM, which sequentially applies channel and spatial attention to highlight informative channels and spatial locations. A residual connection then fuses the refined feature map F'' with the original F to retain multilevel information. Finally, two convolutional downsampling operations progressively reduce the feature dimensionality to $F''' \in R^{C_2 \times W \times W}$. In this process, $T = 5$ denotes the temporal sequence, including two months before and after the target month; $C_1 = 32$ is the number of channels in the intermediate feature map; and $C_2 = 4$ is the dimensionality of the final compressed representation.

2) Multigranularity Feature Extraction of XCO_2: The neighboring XCO_2 observation sequence serves as a critical spatial reference for estimating regional carbon concentrations, and the precision of its spatial correlation modeling directly influences the reliability of the retrieval results. Traditional geographically weighted methods—such as inverse distance weighting (IDW)—construct spatial weight matrices based on predefined distance decay functions. While these approaches can generate a basic concentration field, they often fail to capture the complex nonlinear characteristics of atmospheric transport.

To address the aforementioned challenges, this study develops a multigranularity input strategy for XCO_2 sequence information within the integrated processing pipeline. In this context, granularity refers to the minimal processing unit of XCO_2 information. The strategy builds on three design considerations. First, neighboring points are sampled at multiple granularities—i.e., with varying numbers of nearest observations—while preserving their original spatial distribution characteristics. Second, given the spatial anisotropy of satellite observations (e.g., OCO-2 measurements typically have more samples along the along-track than across-track direction), directional weights are introduced in the horizontal plane to achieve a more isotropic representation of the observation distribution, thereby mitigating potential striping artifacts in the reconstructed outputs. Third, XCO_2 sequence features extracted at different granularities are combined, with each granularity-specific feature assigned a learnable weight optimized via a feedforward neural network, allowing the system to adaptively emphasize the most informative representation. A practical example is presented in Fig. S1 of the supplementary material to illustrate the effectiveness of this module.

Considering computational efficiency, this study adopts five levels of granularity with N = [5, 10, 20, 50, 100] neighboring points. To address spatial anisotropy, three levels of directional weighting in the horizontal direction—1×, 5×, and 10×—are applied. Combining these configurations results in a total of 15 distinct XCO_2 sequences. Fig. 2(c) illustrates the workflow of the multigranularity XCO_2 feature extraction module. Specifically, each XCO_2 sequence is formatted as a 1-D vector of size ($2N$, 1), where satellite observations and their corresponding relative distances are alternately arranged. Each of the 15 input vectors is passed through a dedicated linear layer tailored to its dimensionality, yielding a feature vector of shape (32, 1). These are then compressed via two fully connected (FC) layers into vectors of shape (4, 1). Finally, all 15 vectors are concatenated and processed through another FC layer to produce a single fused XCO_2 feature vector of shape (4, 1), integrating multigranularity and directional information for downstream estimation tasks.

3) Feature Integration: After extracting features from multiple input modalities through the preceding modules, a dedicated regression module is required to effectively integrate these features and produce accurate XCO_2 predictions. As illustrated in Fig. 2(d), we designed a regression architecture that begins by concatenating three inputs along the feature dimension: the auxiliary feature tensor (with shape $C2 \times W \times W$), the neighboring observation vector (4×1), and the spatiotemporal encoding vector (3×1). These inputs are flattened and merged to form a hybrid feature vector $\Phi \in R^{(C_2 \cdot W^2 + 4 + 3)}$.

To enable multiscale feature interaction and nonlinear modeling, a deep residual network is employed. The main path performs regression through a stack of FC layers, while a skip connection is used to directly propagate the neighboring XCO_2 observations, thereby preserving physical constraints inherent in the original measurements [39]. In addition, to improve temporal continuity and leverage prior knowledge, the output of the final FC layer incorporates the XCO_2 value from the same month in the previous year as an auxiliary prior. For the year 2019, where no historical predictions are available, the corresponding CT output is used as a substitute prior. The model is trained and optimized iteratively to yield the final prediction results.

IV. Results and Discussion

A. Validation of Raw Satellite XCO_2 Product Accuracy

This study evaluates the accuracy of individual satellite XCO_2 products—after data screening and 0.05° gridding—by comparing them with ground-based TCCON observations. For each TCCON station, we averaged all valid gridded satellite XCO_2 values within a 100 km radius for each month. Corresponding TCCON monthly means were derived by averaging observations collected within ±1 h of the satellite overpass time. These ST pairs were then matched for validation. Fig. 3 presents scatter plots comparing TCCON observations with satellite-retrieved and fused XCO_2 data. Among the evaluated

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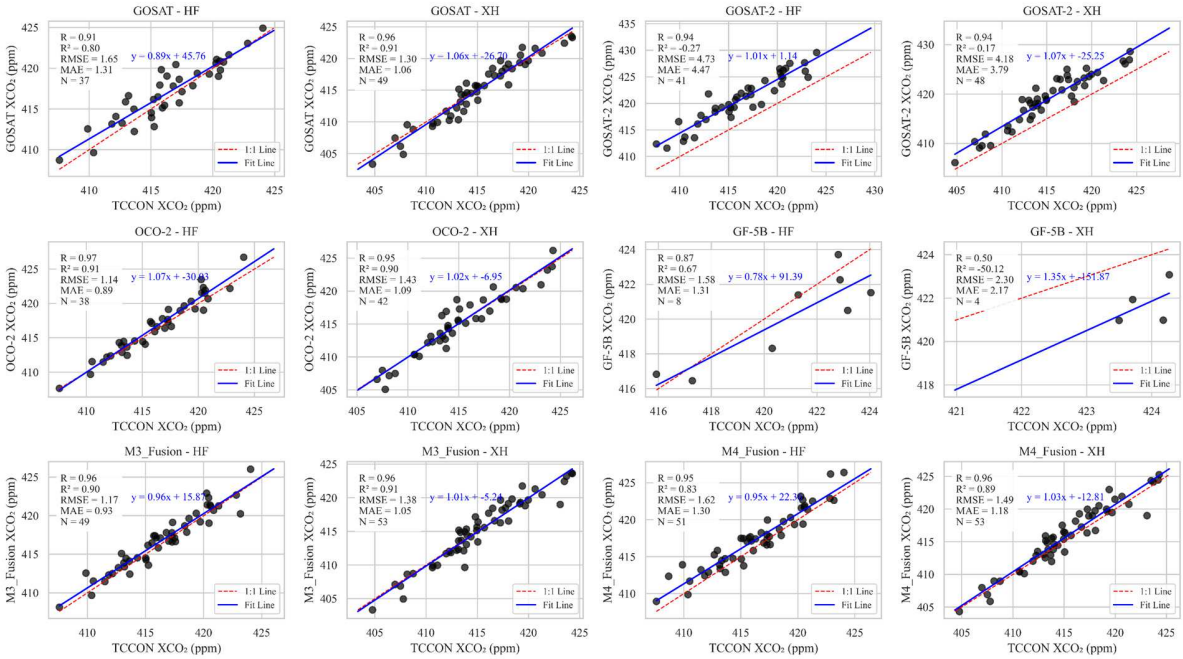


Fig. 3. Validation results of satellite and initial fused XCO_2 evaluated using TCCON XCO_2 measurements.

products, OCO-2 and GOSAT exhibit good agreement with the TCCON measurements, with most data points clustering around the 1:1 reference line and showing no significant overestimation or underestimation. At Xianghe station, the RMSEs for OCO-2, GOSAT, and GF-5B were 1.43, 1.27, and 1.71 ppm, respectively, while their mean absolute errors (MAEs) were 1.09, 1.04, and 1.71 ppm. At Hefei station, RMSEs were 1.14, 1.65, and 1.47 ppm, with corresponding MAEs of 0.89, 1.31, and 1.20 ppm. Notably, the GF-5B product included only data from 2023 to 2024, and TCCON observations at Xianghe were only available from January to May 2023, resulting in a limited number of valid GF-5B data pairs. In contrast, GOSAT-2 exhibited significantly poorer performance. Its RMSE and MAE reached 4.14 and 3.74 ppm at Xianghe, and 4.75 and 4.50 ppm at Hefei, respectively. In addition, GOSAT-2 systematically overestimated XCO_2 levels. This may be attributed to the lack of bias correction in its publicly released product [40].

Although column-averaged bias corrections were applied to each satellite dataset prior to fusion and uncertainty-based weighting was employed during the averaging process, the accuracy of the fused XCO_2 product is inevitably influenced by the inherent precision of the individual input sources. To establish a meaningful baseline for evaluating the model-generated dataset quality, two comparative fusion schemes were developed: 1) M_3_Fusion, which combines GOSAT, OCO-2, and GF-5B—three satellites with relatively high data quality; and 2) M_4_Fusion, which additionally incorporates GOSAT-2, known for its lower accuracy.

For M_3_Fusion, the RMSE and MAE against TCCON observations at Xianghe station were 1.29 and 1.00 ppm, respectively, while at Hefei station, these metrics were 1.17 and 0.93 ppm. Compared to the individual performance of OCO-2, GOSAT, and GF-5B, M_3_Fusion showed a modest degradation in accuracy. This is primarily attributed to the broader inclusion of observations in the fusion process, which increases the sample size and helps mitigate the influence of outliers present in smaller subsets. In contrast, M_4_Fusion exhibited a noticeable drop in accuracy due to the propagation of large errors from the GOSAT-2 dataset into the fusion output. Despite its lower precision, M_4_Fusion was constructed intentionally to test the model's ability to correct for degraded input quality and to evaluate its robustness under challenging observational conditions.

B. Impact of Observational Quality and Spatial Coverage on Model Performance

From the perspective of model inputs, satellite observations serve as the most critical information source. This study investigates how various combinations of satellite datasets affect the quality of the reconstructed XCO_2 maps. Specifically, three fused products are evaluated: 1) M_2_Fusion of OCO-2 and GOSAT; 2) M_3_Fusion of OCO-2, GOSAT, and GF-5B; and 3) M_4_Fusion of OCO-2, GOSAT, GOSAT-2, and GF-5B. These configurations essentially serve as ablation experiments, where the model is trained and validated separately with different combinations of satellite inputs to explicitly examine how the observational quality and spatial coverage of different satellites affect the model performance. Table II presents a comparison of the statistical metrics between the model-predicted monthly mean XCO_2 and TCCON observations under these different satellite input scenarios, as well as the validation results on the temporally partitioned test sets. In addition, Fig. 4 illustrates the annual average XCO_2 distribution maps for 2023 derived from each satellite combination.

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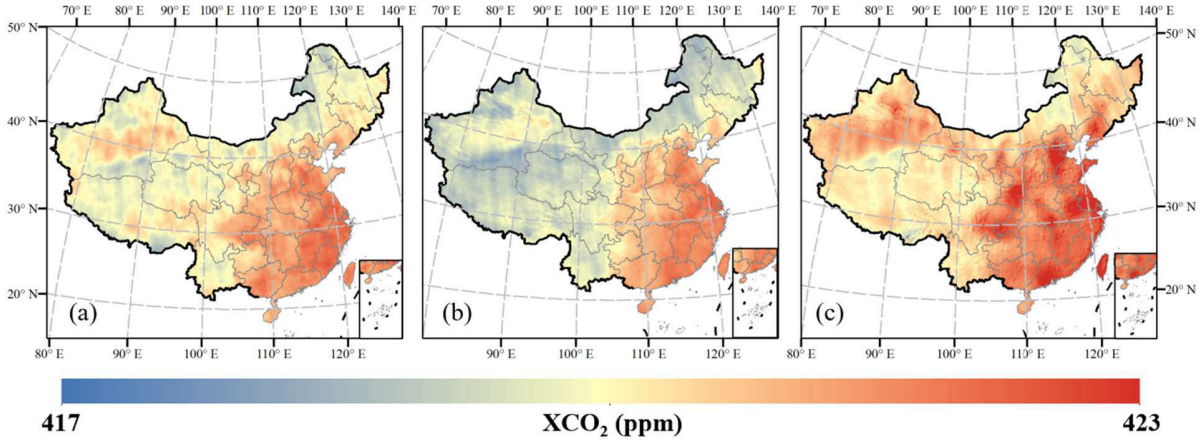


Fig. 4. Average XCO_2 distribution in 2023 derived from (a) M_2, (b) M_3, and (c) M_4.

TABLE II
SUMMARY OF STATISTICAL METRICS

Validation with TCCON					
Data		**r**	**R^2**	**RMSE**	**MAE**
M_2		0.96	0.79	1.94	1.54
M_3		0.96	0.88	1.50	1.16
M_4		0.94	0.66	2.48	2.07
Ablation Comparison					
Period	**Data**	**r**	**R^2**	**RMSE**	**MAE**
2019-2023	M_2	0.96	0.81	1.84	1.47
	M_3	0.96	0.87	1.44	1.21
2023-2024	M_2	0.96	0.81	1.80	1.50
	M_3	0.96	0.91	1.34	1.14

These results enable a comprehensive assessment of the trade-offs between observational diversity and data quality in the context of multisatellite fusion-based XCO_2 estimation.

As shown in Table II, the performance metrics of the M_2-based reconstruction yield the R^2 of 0.79, an RMSE of 1.94 ppm, and an MAE of 1.54 ppm. In comparison, the M_3 product achieves improved results, with the R^2 of 0.88, an RMSE of 1.50 ppm, and an MAE of 1.16 ppm. For contrast, an additional fusion scenario (M_4) was created by integrating GOSAT-2 data—known for its lower accuracy—on top of M_3. This leads to a performance degradation. The R^2 drops to 0.66, while RMSE and MAE increase to 2.48 and 2.07 ppm, respectively. To further evaluate the robustness of GF-5B, we conducted ablation experiments by separating the validation into two periods (from 2019 to 2022 and from 2023 to 2024). From 2019 to 2022, the model trained with the M_3 dataset, which includes GF-5B, achieved a correlation of 0.96, a coefficient of determination (R^2) of 0.87, an RMSE of 1.44 ppm, and an MAE of 1.21 ppm. In contrast, the M_2 configuration without GF-5B yielded the same correlation of 0.96 but a lower R^2 of 0.81, along with larger errors (RMSE = 1.84 ppm and MAE = 1.47 ppm). A similar pattern was observed from 2023 to 2024, where the M_3-trained model reached the R^2 of 0.91, an RMSE of 1.34 ppm, and an MAE of 1.14 ppm, outperforming M_2, which again remained at R^2 = 0.81 with higher errors (RMSE = 1.80 ppm and MAE = 1.50 ppm). These results demonstrate that the inclusion of GF-5B consistently improves model accuracy across both time periods, confirming that its contribution is robust rather than limited to its coverage window.

Although the independent validation against satellite products does not provide direct evidence that GF-5B measurements are more accurate than those of OCO-2 or GOSAT, incorporating GF-5B into the training process nevertheless enhances the overall model accuracy. This improvement can be attributed to two factors. First, the comparison with M_4 indicates that the inclusion of lower quality data degrades model performance, whereas the positive effect of GF-5B implies that its measurements are of reasonably high quality. Second, GF-5B offers denser and more spatially uniform coverage, particularly in regions that are poorly sampled by OCO-2 or GOSAT. These additional observations enrich the training dataset and allow the model to learn more representative spatiotemporal patterns, thereby improving reconstruction accuracy.

The spatial distribution patterns presented in Fig. 4 provide a more intuitive understanding of how different satellite inputs influence the reconstruction results. Fig. 4(a) illustrates the annual mean XCO_2 distribution derived from M_2, which only incorporates OCO-2 and GOSAT observations. Due to the relatively limited number of original satellite measurements, the reconstruction appears spatially coarse, particularly in

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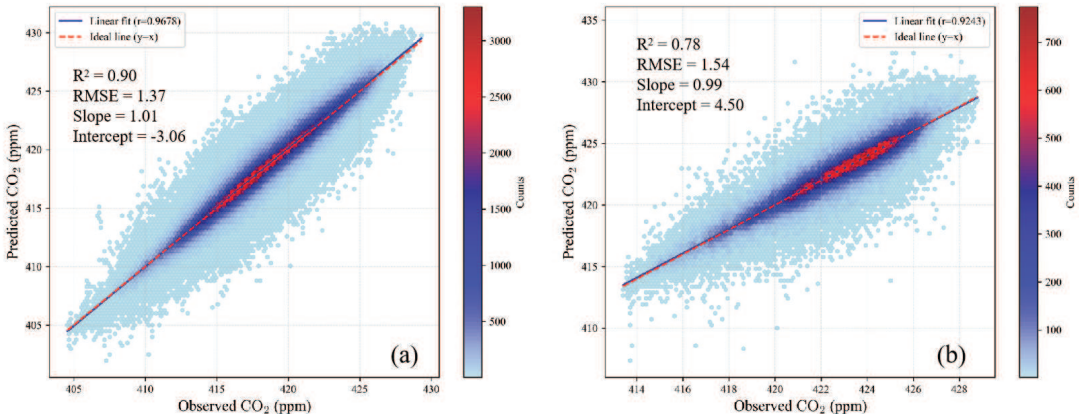


Fig. 5. (a) Overall performance evaluation. (b) Temporal holdout experiment.

transitional zones such as the Taklamakan Desert region in Xinjiang, where sharp gradients are observed instead of the smoother patterns seen in M_3 and M_4. In contrast, Fig. 4(b) demonstrates that the inclusion of GF-5B in the M_3 product effectively mitigates the overestimation of XCO_2 observed in the western regions of China in M_2. The transition from high to low values becomes more reasonable and spatially consistent. Fig. 4(c) shows the reconstruction based on M_4, which includes additional GOSAT-2 data. As the number of observations increases, finer spatial details emerge, enabling clearer identification of high-emission regions such as Chongqing, Urumqi, and Beijing–Tianjin–Hebei area. However, due to the known overestimation bias in GOSAT-2 observations, the overall XCO_2 levels in M_4 appear to be systematically elevated. This suggests that applying bias correction to GOSAT-2 data in future work could further enhance the accuracy and reliability of the reconstruction. We further discuss this issue in the supplementary material.

C. Validation With Satellite Observation

1) Overall Performance Evaluation: To rigorously assess the model's performance and mitigate potential bias from a single data split, we employed a tenfold cross-validation strategy. In this setting, the dataset was partitioned into ten equal subsets, with each subset used once as the validation set, while the remaining nine subsets served as the training set. The evaluation metrics were then averaged across all folds to ensure robustness.

As shown in Fig. 5(a), the cross-validation results, obtained by aggregating predictions from all folds, indicate that the model achieves high overall accuracy. Specifically, the Pearson correlation coefficient (R) for the combined validation set is 0.98, with a coefficient of determination (R^2) of 0.90. The RMSE and MAE are 1.37 and 0.31 ppm, respectively. Linear regression between the aggregated predicted and observed values yields a slope of 1.01 and an intercept of approximately −3.06 ppm. These results demonstrate that the model effectively captures the underlying relationships between input features and satellite-based XCO_2 observations, providing robust performance across the entire dataset.

2) Temporal Holdout Experiment: In addition to the cross-validation experiment, we conducted extrapolation evaluations to examine the model's generalization capability beyond the training domain. In this setting, we trained the model using data from 2022 to 2023 and validated it with independent observations from 2024. As shown in Fig. 5(b), the extrapolation experiment yielded the R^2 of 0.78, an RMSE of 1.54 ppm, and a regression slope and intercept of 0.99 and 4.5 ppm, respectively. Compared with the overall cross-validation performance, the extrapolation results are relatively weaker. However, the model still demonstrates a satisfactory predictive capability under unseen temporal conditions.

D. Validation With Site Base

To align with satellite overpass times, we selected TCCON observations recorded between 12:30 and 14:30 for comparison with the model predictions. The results are illustrated in Fig. 6. To further quantify the model's accuracy, we calculated the monthly average of TCCON observations and evaluated the RMSE and Pearson correlation coefficient (R) between the model predictions and TCCON data within varying radii from the station. For benchmarking purposes, the performance of the M_3_Fusion dataset is also included. The statistical metrics under different spatial thresholds are summarized in Table III.

Fig. 6 demonstrates that the model predictions align closely with the TCCON observations and exhibit clear seasonal variability. Due to its southern location, Hefei site has significantly fewer valid satellite observations compared to Xianghe site, leading to relatively lower prediction accuracy at Hefei. As shown in Table III, the model accuracy improves to some extent as the search radius increases. At Xianghe site, the coefficient of determination (R^2) and RMSE for a 5 km radius decrease by only 0.02 and 0.22 ppm, respectively, compared to the 100 km radius, indicating good model stability. Furthermore, comparison with the original M_3_Fusion grids reveals that the model effectively corrects outliers in the raw satellite observations. At the 100 km radius, the model's RMSE outperforms the M_3_Fusion result by 0.04 ppm. This improvement becomes more significant with smaller search radii. For instance, at a 20 km radius, the RMSE is reduced from 2.08 (M_3) to 1.43 ppm by the model.

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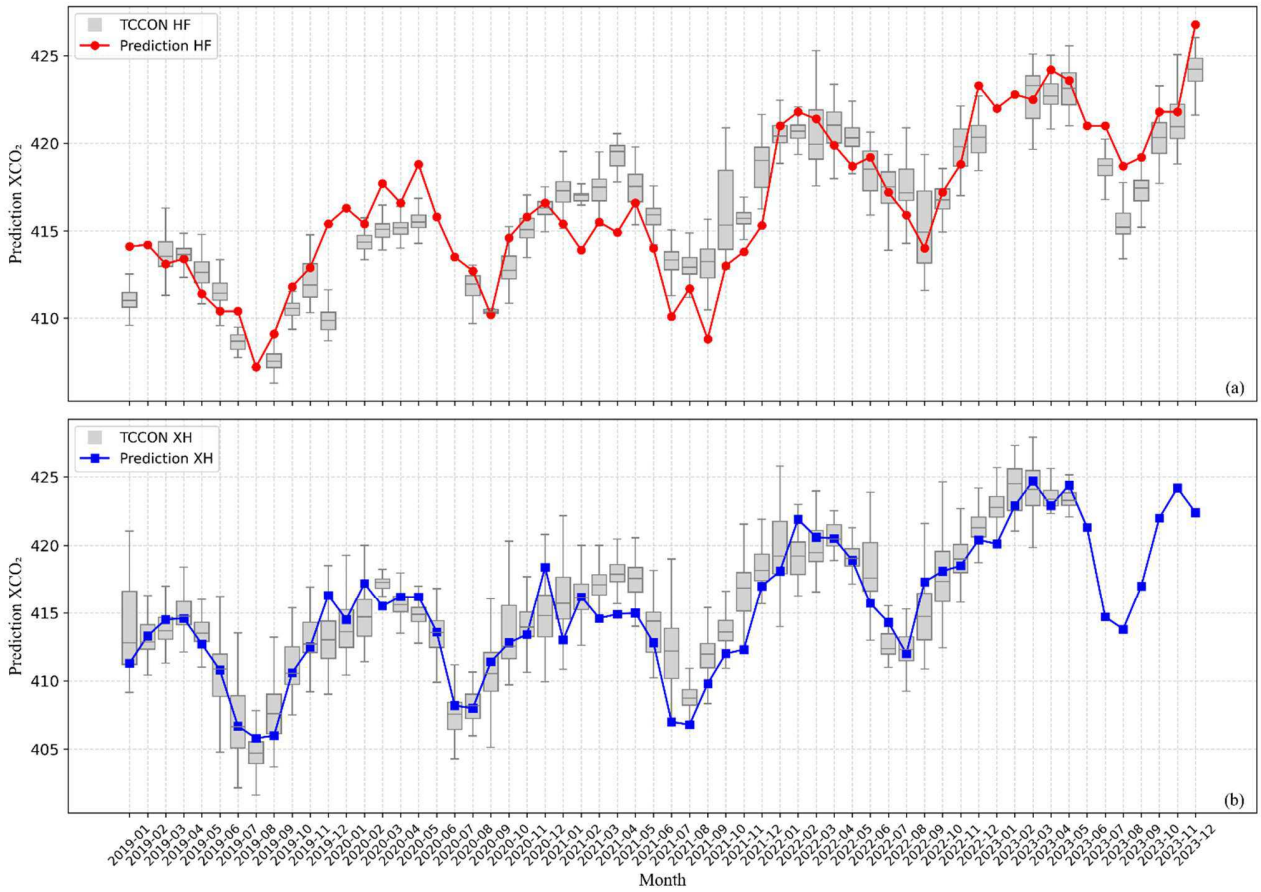


Fig. 6. Comparison between predicted XCO_2 and TCCON observations at (a) HF and (b) XH sites.

TABLE III
COMPARISON OF R^2 AND RMSE BETWEEN MODEL ESTIMATES AND M_3

Station	Radius(km)	R^2	M_3 R^2	RMSE	M_3 RMSE
HF	5	0.86	-	1.47	-
	10	0.86	-	1.45	-
	20	0.87	0.95	1.52	0.82
	50	0.88	0.89	1.38	1.17
	100	0.89	0.90	1.32	1.17
XH	5	0.88	-	1.57	-
	10	0.88	0.78	1.56	1.93
	20	0.89	0.78	1.43	2.08
	50	0.91	0.87	1.37	1.58
	100	0.92	0.91	1.25	1.38

E. Comparison With Assimilated Data

Fig. 7(a)–(c) presents the spatial distributions of the average XCO_2 concentrations from the proposed model, CT, and CAMS for the years 2019 and 2020. Overall, the model results exhibit spatial patterns similar to those of CT and CAMS. Fig. 7(d) and (e) shows the spatial distribution of differences between the model outputs and CT and CAMS, respectively, while Fig. 7(g) illustrates the histogram of these differences.

Specifically, the model tends to overestimate XCO_2 in regions with relatively low anthropogenic emissions in western China, and underestimate concentrations in the eastern regions. When compared to CT, the model yields a mean difference of −0.28 ppm with a standard deviation of 0.53 ppm. Positive differences account for 31.8% of the data, with a maximum positive bias of 1.38 ppm, primarily located in the Taklamakan Desert, the Qinghai–Tibet Plateau, and the northwestern part of Sichuan Province. Negative differences account for 68.2%,

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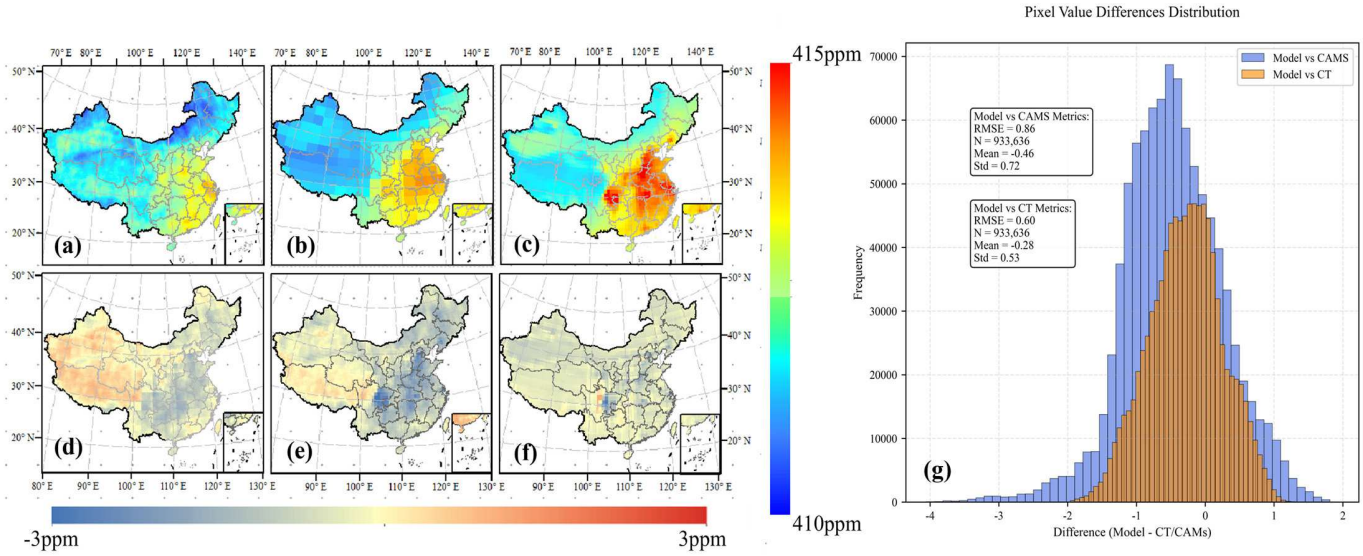


Fig. 7. Average XCO_2 from 2019 to 2020 for (a) model result of this study, (b) CT, and (c) CAMS. Differences of (d) model minus CT, (e) model minus CAMS, (f) CT minus CAMS, and (g) distribution of pixel value differences between the model and the CT and CAMS.

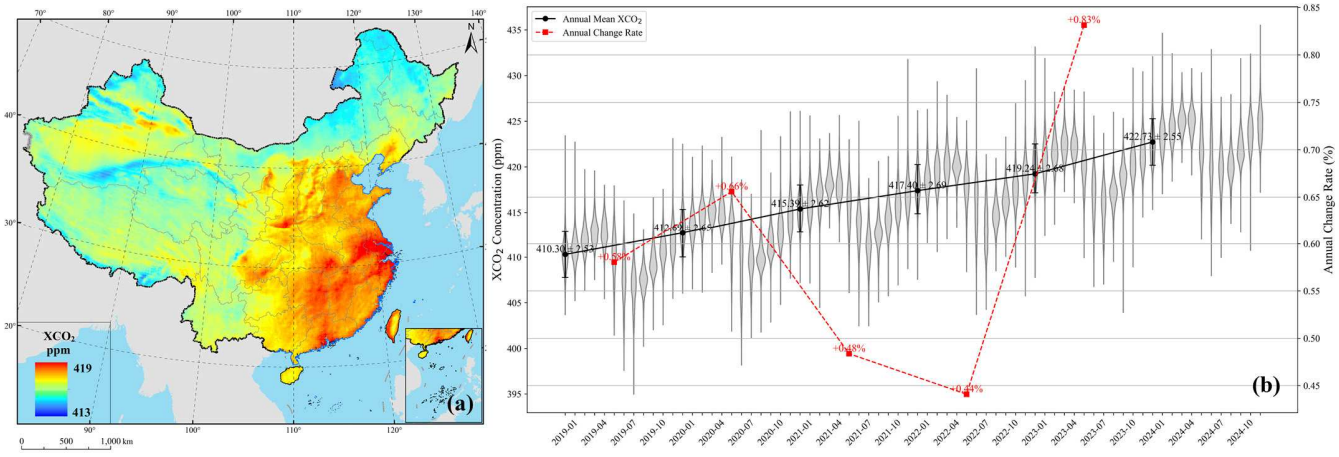


Fig. 8. (a) Average XCO_2 from 2019 to 2024. (b) Algorithm-generated monthly XCO_2 pixel values and corresponding interannual variations.

with a maximum negative bias of −2.03 ppm, concentrated in the southern Beijing–Tianjin–Hebei region, Central China, and eastern Sichuan. Compared to CAMS, the model shows a mean difference of −0.46 ppm and a standard deviation of 0.72 ppm. The maximum difference reaches 4.2 ppm in the eastern Sichuan Basin, and negative differences around −1 ppm are commonly observed across Central China, indicating a more pronounced underestimation. This discrepancy may be attributed to the relatively sparse valid satellite observations over these regions.

F. Interannual Variability

Fig. 8(a) illustrates the averaged XCO_2 distributions reconstructed from 2019 to 2024. Spatially, atmospheric CO_2 concentrations in China generally exhibit a gradient of low values in the northwest and high values in the southeast. Major high-emission regions are primarily located in the Fenwei Plain, the Yangtze River Delta, and the Pearl River Delta. Taking 2023 as an example, the national annual mean XCO_2 concentration was 419.24 ppm. The corresponding values in the aforementioned high-emission regions were 420.81, 422.76, and 421.85 ppm, all exceeding the national average.

Fig. 8(b) presents the monthly gridded XCO_2 values and interannual variation trends derived from the proposed algorithm. Overall, the results exhibit a clear seasonal cycle. XCO_2 concentrations typically peak in April, decrease from May to August, and then increase steadily from September to the following April. The annual mean concentration increased from 410.30 ppm in 2019 to 422.73 ppm in 2024, corresponding to an average annual growth rate of approximately 2–3 ppm. The line chart of interannual growth rates indicates fluctuations between 0.4% and 0.9%, with the most significant increase observed between 2023 and 2024, reaching 0.83%.

G. Competitive Baselines and Model Comparison

To further evaluate the performance of our model, we introduced a set of competitive baselines, including both

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TABLE IV
CONTROLLED BENCHMARKING

Model	R^2	RMSE	MAE
Ours	0.88	1.79	1.32
DF	0.88	1.83	1.37
RF	0.85	2.05	1.54
XGBOOST	0.86	1.99	1.47

INDIRECT LITERATURE-BASED COMPARISON

Model	10-fold-cross-validation result		Validation with TCCON		Spatial resolution(°)	Temporal resolution	Real time	referance
	R^2	RMSE(ppm)	R^2	RMSE(ppm)				
Ours	0.90	1.37	0.88	1.50	0.05	Monthly	√	-
DF	0.90	2.18	0.81	1.77/2.01*	0.05	Monthly	×	[10]
DSTM	0.99	0.33	0.86	2.67	0.1	Daily	×	[20]
CNN-LSTM	-	-	-	0.93/1.29*	0.25	Monthly	×	[18]

Note*: The values in the table represent validation results, where the number before the slash (/) indicates the accuracy at Hefei station, and the number after the slash (/) indicates the accuracy at Xianghe station.

recent state-of-the-art methods and classical models. However, a direct head-to-head benchmark is inherently challenging because these models rely on different input formats, preprocessing pipelines, and temporal–spatial resolutions. To address this, we adopted a two-pronged strategy.

1) Controlled Benchmarking: We selected several open source and widely used classical models. For fairness, we constrained the comparison to identical spatial resolution and temporal coverage, and adjusted input variables where possible. This approach allowed us to approximate a head-to-head benchmark by minimizing differences in data preprocessing.

As shown in Table IV, our model consistently achieved the best performance in both cross validation and independent TCCON validation, with the lowest errors (RMSE = 1.79/1.42 ppm) and the highest R^2 values (0.88/0.87). In comparison, deep forset (DF) achieved comparable accuracy in cross validation but declined markedly against TCCON, while random forest (RF) and extreme gradient boosting (XGBoost) performed substantially worse. These results highlight the superior robustness and generalization capability of our framework.

2) Indirect Literature-Based Comparison: We collected key performance metrics reported in recent studies of state-of-the-art XCO_2 reconstruction models and compared them with our framework. In addition to quantitative indicators such as RMSE and R^2, we also conducted a qualitative comparison of each model's spatial and temporal resolution and its capability to generate near-real-time XCO_2 products. This comparison highlights the relative advantages of our approach in terms of both reconstruction accuracy and operational applicability.

As shown in Table IV, our framework exhibits competitive or superior accuracy compared with recent literature. In terms of cross validation, our method achieves the R^2 of 0.90 and an RMSE of 1.37 ppm, significantly outperforming the DF baseline (RMSE = 2.18 ppm) and approaching the performance of advanced spatiotemporal fusion models such as DSTM. When validated against independent TCCON measurements, our model maintains high accuracy (R^2 = 0.88 and RMSE = 1.50 ppm), while DF and DSTM show reduced generalization with larger RMSE values (1.77–2.67 ppm). The CNN–LSTM framework reported in previous studies demonstrates competitive validation RMSE values (0.93–1.29 ppm), but its coarse spatial resolution (0.25°) and lack of real-time capability limit its operational applicability. In contrast, our method achieves a finer spatial resolution of 0.05° with near-real-time capability, highlighting a favorable balance between reconstruction accuracy, spatial detail, and practical usability.

H. Limitations and Practical Considerations

While the proposed integrated pipeline enables near-real-time reconstruction of XCO_2 at high spatial resolution over China, several practical considerations remain. The end-to-end processing latency is primarily determined by the availability of satellite and auxiliary datasets. As of August 2025, most required datasets are available within one month of acquisition, whereas nighttime light data exhibit a delay of approximately two months. Computationally, model training was performed on an AMD 5955WX processor with a PTX4090 GPU, requiring roughly 8 h for 200 epochs, reflecting the substantial cost of multisource data fusion and spatiotemporal attention mechanisms. Coverage gaps in satellite observations, particularly for GOSAT and OCO-2 during summer over southern China, result in locally reduced model performance. Nonetheless, the overall reconstructed fields remain reliable. In addition, regions with sparse or missing observations may introduce additional uncertainty, though these effects are mitigated through the model's feature-fusion-aware and granularity-aware mechanisms. Finally, while the pipeline is tailored to China, its design—integrating multiple satellite sources and

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auxiliary variables—provides a flexible framework potentially extendable to other regions, though careful adaptation would be required to account for differences in satellite coverage, temporal resolution, and local atmospheric conditions. These considerations collectively inform both the current limitations and the practical applicability of the pipeline for near-real-time carbon monitoring.

V. Conclusion

This study develops an integrated processing pipeline that refines and combines several classical deep learning modules within a multigranularity spatiotemporal attention framework to reconstruct continuous XCO_2 distributions over China at a spatial resolution of 0.05°. The pipeline fuses multisource satellite observations—including GOSAT, GOSAT-2, OCO-2, and the underexplored GF-5B sensor—with auxiliary variables such as NDVI, EVI, NO_2 concentrations, nighttime light data, and ERA5 meteorological parameters. This configuration enables the deep extraction and fused representation of multiscale features while leveraging the complementary strengths of diverse datasets for enhanced reconstruction quality.

The main contributions of this work are threefold. First, the processing pipeline is designed to operate entirely on satellite-derived observations, without reliance on lagged model products (e.g., CT or CAMS), thereby overcoming the considerable time delay associated with model simulation data and demonstrating the feasibility of near-real-time carbon monitoring at high spatial resolution. Second, the framework integrates multisource satellite observations—including GOSAT, GOSAT-2, OCO-2, and the first time underexplored GF-5B sensor—together with auxiliary datasets such as NDVI, EVI, NO_2 concentrations, nighttime light, and ERA5 meteorological parameters, leveraging the complementary strengths of diverse data sources and indirectly confirming the high quality of GF-5B observations. Third, the pipeline incorporates an improved TSCAM module with multigranularity modeling capability and spatial anisotropy-aware mechanisms, along with a neighboring observation submodule, to enhance spatial consistency and stability in reconstructed XCO_2 fields. Experimental results demonstrate that the proposed model performs well across multiple validation levels. Compared with TCCON ground-based observations, the predicted values accurately capture the temporal variation patterns of atmospheric CO_2 concentrations. The model also exhibits strong bias correction capability and robustness in data-sparse regions. Comparisons with data assimilation products such as CT and CAMS further confirm the spatial consistency and reliability of the model outputs. Interannual variation analysis shows that the model is sensitive to both the overall growth trend and regional disparities in CO_2 concentrations.

This study explores the feasibility of reconstructing near-real-time, high-spatial-resolution XCO_2 distributions based on multisource satellite observations. Future work will consider the integration of higher frequency remote sensing data (e.g., DaQi (DQ)-1) to further enhance the model's performance.

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Ruizhi Chen received the B.S. degree from Southwest University, Chongqing, China, in 2023. He is currently pursuing the M.S. degree with Chinese Research Academy of Environmental Sciences, Beijing, China.

His research interests include atmospheric remote sensing and CO_2 concentration mapping.



Mingmin Zou received the Ph.D. degree in cartography and geographical information systems from the University of Chinese Academy of Sciences, Beijing, China, in 2013.

He is currently a Professor with the Institute of Physical Science and Information Technology, Anhui University, Hefei, China. His work focuses on new developments in remote sensing of atmospheric trace gases and environmental applications.



Cuihong Chen received the master's degree from Tsinghua University, Beijing, China, in 2018.

Currently, she works at the Satellite Application Center for Ecology and Environment, Ministry of Ecology and Environment, Beijing. Her research interests include remote sensing monitoring and assessment of greenhouse gases.



Huizhen Xie received the master's degree from China University of Mining and Technology, Beijing, China, in 2023.

Currently, she works at the Satellite Application Center for Ecology and Environment, Ministry of Ecology and Environment, Beijing.



Chunyan Zhou received the Ph.D. degree from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, China, in 2009.

She is a Senior Engineer at the Satellite Application Center for Ecology and Environment, Ministry of Ecology and Environment, Beijing. Her research interests include remote sensing monitoring and the application of aerosols, trace gases, and greenhouse gases.



Huiqin Mao received the Ph.D. degree from the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, in 2010.

Currently, she works at the Satellite Application Center for Ecology and Environment, Ministry of Ecology and Environment, Beijing. Her research focuses on the application of satellite remote sensing technology in addressing climate change.



Zhongting Wang received the Ph.D. degree in cartography and geographical information systems from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, China, in 2008.

From 2008 to 2024, he was a Senior Engineer with the Satellite Application Center for Ecology and Environment, Ministry of Ecology and Environment, Beijing. Currently, he is a Senior Engineer with North China University of Water Resources and Electric Power, Zhengzhou, China. His primary research interests include aerosol and $PM_{2.5}$ retrieval from satellite data, CO_2 monitoring with satellite data, and atmospheric radiative transfer.

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